CPSC 3720 - Assignment #3

Due Date: November 7, 2025

Total Grades: 50

Assignment #3 – Software Reuse and Project

For Assignment #3, you have the option of completing either an individual or a team project (maximum 3 students) on any of the topics covered in Chapter 15 (Software Reuse).

Use of Tools and Resources – Guidelines

Students/Teams are allowed to use publicly available datasets (e.g., Kaggle, City of Calgary, NOAA, weather data or any other data set – you can ask help from instructor for any specific data sets if needed) or any specific datasets relevant to their project topic. You may also use AI generative tools (such as ChatGPT, GitHub Copilot, etc.) as support tools to assist with planning, coding, and documentation. However:

• You must not copy code directly from any source.

• You must cite all tools, datasets, and references used.

• AI tools should be used to support your learning and development, not replace original work.

Theory Questions (Each question is worth 5 marks)

Total Grades: 10

1. Question 1: Give five circumstances where you might recommend against software reuse.

2. Question 2: Why do many large organizations adopt ERP systems for their information infrastructure? What reuse-related challenges might arise during ERP deployment?

Project: Reusable Data Visualization Dashboard Framework for Environmental Data

Total Grades: 40

Course: CPSC 3720 - Software Engineering

Instructor: Dr. Jaspreet Kaur

Maximum pages allowed: 6 for the project part

Abstract

This project focuses on developing a reusable data visualization dashboard framework that supports modular integration of different datasets and visualization types. The goal is to

demonstrate software reuse in component-based software engineering by allowing flexible and maintainable visualization tools for environmental data analysis.

1. Motivation

Environmental data visualization is often dataset-specific, resulting in limited reusability. This project provides a framework for flexible visualization, saving development time and promoting sustainable reuse.

2. Objectives

• Develop a reusable visualization framework.
• Demonstrate modular design.
• Implement visual components.
• Enable data interchangeability.

3. Literature Review and Background

Krueger (1992) emphasized software reuse efficiency. Gamma et al. (1994) introduced design patterns that improve reusability. Sommerville (2020) outlined architectural reuse benefits. This project builds upon these principles by applying component reuse to data visualization.

4. Software Requirements Specification (SRS)

Functional Requirements:
• FR1: Load CSV/JSON datasets.
• FR2: Allow parameter selection.
• FR3: Provide multiple visualization types.

Non-Functional Requirements:
• Performance: Load under 2 seconds.
• Reusability: 80% component reuse.

5. Methodology

• Literature review and requirements gathering
• System design and architecture
• Data loader implementation and Visualization components
• Dashboard integration (if any)
• Testing and Results and documentation

6. System Design

The system follows a modular architecture with reusable components for data loading, filtering, and visualization. Libraries used include Chart.js and Plotly for visualization, and React.js for front-end management.

7. Dataset

You can use any open or specific datasets. Please cite all resources.

8. Testing

Include Unit, Integration, Usability, and Performance testing. Example: TC01 – Load valid CSV → expected successful visualization.

9. Results

Describe the outcomes of your implementation including performance metrics and component reuse.

10. Conclusions

Please mention conclusions specific to your project/application. Sample: Software reuse significantly improves system scalability and maintainability. The developed framework successfully demonstrates modular, reusable design principles applicable in environmental data research. What worked and what did not work.

References

• Krueger, C. W. (1992). Software reuse. ACM Computing Surveys.
• Gamma, E., Helm, R., Johnson, R., & Vlissides, J. (1994). Design Patterns.
• Sommerville, I. (2020). Software Engineering (10th Ed.).
• Kaur, J., & Singh, G. (2023). Reusable Data Visualization Components for Educational Tools.